

Pandemic-Period Comparison

Psycholinguistic markers pre / during / post pandemic
after Yahya & Abdul Rahim (2023), with per-user rank-based statistics

Dataset: New Data Only

Section 1 — Diagnostics

Period Definitions

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pre      : PostDate < 2020-03-11
during   : 2020-03-11 ≤ PostDate < 2022-03-23
post     : PostDate ≥ 2022-03-23
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Cutoff = WHO pandemic declaration (2020-03-11).

End = end of most Dutch COVID restrictions (2022-03-23).

Dataset Variant × Pandemic Period — messages and unique users
⚠ **Period effects are confounded with dataset variant (old export ≈ pre+during, new export ≈ post); yellow = near-empty cell**

variant	period	messages	users
old	pre	4505	169
old	during	11102	291
old	post	2943	96
new_only	pre	94	15
new_only	during	318	23
new_only	post	25092	520
combined	pre	4865	177
combined	during	11960	307
combined	post	6592	247

Users by Number of Periods Appeared In
Users in >1 period violate between-group independence — see the single-period sensitivity analysis in Section 4

periods_appeared_in	n_users	pct_users
1	508	95.8
2	16	3.0
3	6	1.1

Section 2 — Dictionary Introspection

Yahya & Abdul Rahim (2023) §2.3 Categories vs Available Dictionaries
Only available categories are analyzed; '—' = genuinely absent from that source (omission, not a bug)

yahya_group	liwc2015	liwc22	dic_crosscheck
pronouns	i, ipron, ppron, pronoun, shehe, they, we, you	i, ipron, ppron, pronoun, shehe, they, we, you	i, ipron, ppron, pronoun, shehe, they, we, you
function_words	article, auxverb, conj, function, negate, number, prep, quant	article, auxverb, conj, function, negate, number, prep, quant	article, auxverb, conj, function, negate, number, prep, quant
other_grammar	adj, adverb, compare, interrog, verb	adj, adverb, compare, interrog, verb	adj, adverb, compare, interrog, verb
tense	focusfuture, focuspast, focuspresent	focusfuture, focuspast, focuspresent	focusfuture, focuspast, focuspresent
emotion	affect, anger, anx, negemo, posemo, sad	affect, anger, anx, negemo, posemo, sad	affect, anger, anx, negemo, posemo, sad
informal	assent, filler, informal, netspeak, nonflu, swear	assent, filler, informal, netspeak, nonflu, swear	assent, filler, informal, netspeak, nonflu, swear
summary	—	—	—
absolutist	absolutist	—	—

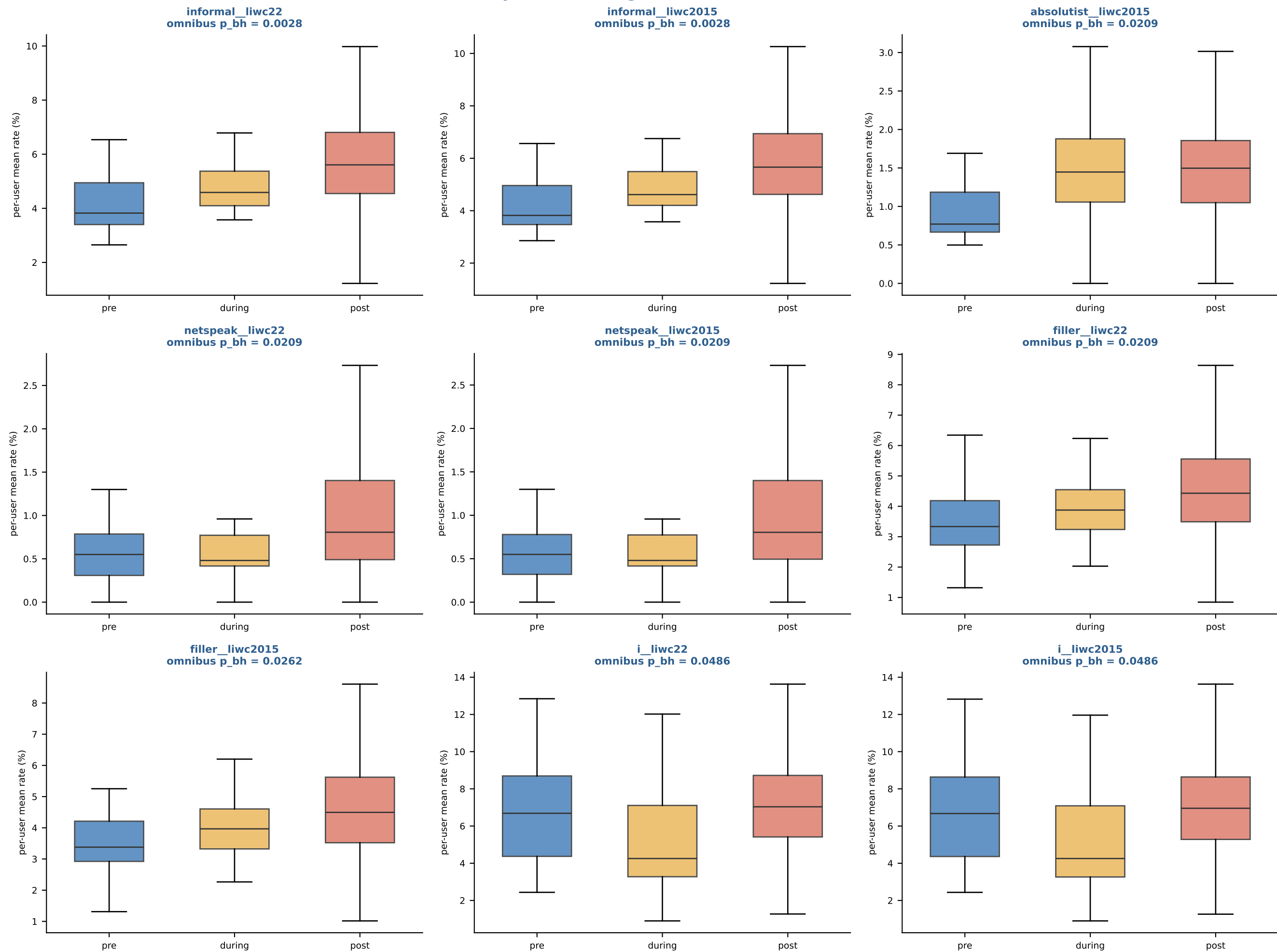
Section 3 — Main Results (all users)

Kruskal-Wallis Omnibus per Feature (per-user rates) effect_size = epsilon-squared; p_bh = BH over pooled omnibus + post-hoc family; yellow = significant										
feature	n_pre	n_during	n_post	comparison	statistic	p_raw	effect_size	note	p_bh	significant
informal_liwc22	15.0	23.0	520.0	omnibus	20.198	4.1e-05	0.0363		0.002812	True
informal_liwc2015	15.0	23.0	520.0	omnibus	19.4482	6e-05	0.0349		0.002812	True
absolutist_liwc2015	15.0	23.0	520.0	omnibus	11.9202	0.00258	0.0214		0.020863	True
netspeak_liwc22	15.0	23.0	520.0	omnibus	12.6004	0.001836	0.0226		0.020863	True
netspeak_liwc2015	15.0	23.0	520.0	omnibus	12.9307	0.001556	0.0232		0.020863	True
filler_liwc22	15.0	23.0	520.0	omnibus	11.9039	0.002601	0.0214		0.020863	True
filler_liwc2015	15.0	23.0	520.0	omnibus	11.0912	0.003905	0.0199		0.026217	True
i_liwc22	15.0	23.0	520.0	omnibus	9.2881	0.009619	0.0167		0.048619	True
i_liwc2015	15.0	23.0	520.0	omnibus	9.2452	0.009827	0.0166		0.048619	True
they_liwc2015	15.0	23.0	520.0	omnibus	6.413	0.040498	0.0115		0.16537	False
sad_liwc22	15.0	23.0	520.0	omnibus	6.0941	0.0475	0.0109		0.16537	False
sad_liwc2015	15.0	23.0	520.0	omnibus	6.1533	0.046114	0.011		0.16537	False
they_liwc22	15.0	23.0	520.0	omnibus	6.4345	0.040065	0.0116		0.16537	False
negemo_liwc2015	15.0	23.0	520.0	omnibus	6.0952	0.047474	0.0109		0.16537	False
negemo_liwc22	15.0	23.0	520.0	omnibus	5.9544	0.050936	0.0107		0.170999	False
you_liwc2015	15.0	23.0	520.0	omnibus	5.0497	0.08007	0.0091		0.217197	False
you_liwc22	15.0	23.0	520.0	omnibus	5.1115	0.077634	0.0092		0.217197	False
prep_liwc2015	15.0	23.0	520.0	omnibus	4.9876	0.082594	0.009		0.217197	False
ipron_liwc22	15.0	23.0	520.0	omnibus	5.0023	0.081991	0.009		0.217197	False
prep_liwc22	15.0	23.0	520.0	omnibus	5.0906	0.07845	0.0091		0.217197	False
adj_liwc2015	15.0	23.0	520.0	omnibus	4.9492	0.084198	0.0089		0.217197	False
ipron_liwc2015	15.0	23.0	520.0	omnibus	4.9187	0.085492	0.0088		0.217197	False
nonflu_liwc2015	15.0	23.0	520.0	omnibus	5.2843	0.071208	0.0095		0.217197	False
adj_liwc22	15.0	23.0	520.0	omnibus	4.7392	0.093516	0.0085		0.219763	False
we_liwc2015	15.0	23.0	520.0	omnibus	4.747	0.093152	0.0085		0.219763	False
we_liwc22	15.0	23.0	520.0	omnibus	4.7592	0.092589	0.0085		0.219763	False
pronoun_liwc2015	15.0	23.0	520.0	omnibus	4.4844	0.106225	0.0081		0.243541	False
pronoun_liwc22	15.0	23.0	520.0	omnibus	4.4327	0.109006	0.008		0.243966	False
shehe_liwc2015	15.0	23.0	520.0	omnibus	3.8922	0.142827	0.007		0.303082	False
shehe_liwc22	15.0	23.0	520.0	omnibus	3.8608	0.145092	0.0069		0.303082	False
focuspast_liwc2015	15.0	23.0	520.0	omnibus	3.5761	0.167283	0.0064		0.328197	False
focuspast_liwc22	15.0	23.0	520.0	omnibus	3.5312	0.171082	0.0063		0.328197	False
nonflu_liwc22	15.0	23.0	520.0	omnibus	3.2503	0.196885	0.0058		0.362886	False
assent_liwc22	15.0	23.0	520.0	omnibus	3.1495	0.20706	0.0057		0.374301	False
quant_liwc2015	15.0	23.0	520.0	omnibus	3.0445	0.218224	0.0055		0.387039	False
quant_liwc22	15.0	23.0	520.0	omnibus	2.9826	0.225082	0.0054		0.39181	False
adverb_liwc22	15.0	23.0	520.0	omnibus	2.816	0.244636	0.0051		0.410638	False
adverb_liwc2015	15.0	23.0	520.0	omnibus	2.8235	0.243714	0.0051		0.410638	False
function_liwc22	15.0	23.0	520.0	omnibus	2.7701	0.250314	0.005		0.412799	False
function_liwc2015	15.0	23.0	520.0	omnibus	2.6858	0.261084	0.0048		0.423136	False
assent_liwc2015	15.0	23.0	520.0	omnibus	2.5362	0.281363	0.0046		0.448273	False
posemo_liwc22	15.0	23.0	520.0	omnibus	2.4788	0.289561	0.0045		0.453645	False
posemo_liwc2015	15.0	23.0	520.0	omnibus	2.4028	0.300779	0.0043		0.463495	False
article_liwc22	15.0	23.0	520.0	omnibus	2.2727	0.32099	0.0041		0.486662	False
article_liwc2015	15.0	23.0	520.0	omnibus	2.1718	0.337599	0.0039		0.50372	False
anger_liwc22	15.0	23.0	520.0	omnibus	2.0541	0.358056	0.0037		0.517805	False
anger_liwc2015	15.0	23.0	520.0	omnibus	2.076	0.354158	0.0037		0.517805	False
focuspresent_liwc2015	15.0	23.0	520.0	omnibus	1.5778	0.454352	0.0028		0.639541	False
focuspresent_liwc22	15.0	23.0	520.0	omnibus	1.5712	0.455843	0.0028		0.639541	False
swear_liwc2015	15.0	23.0	520.0	omnibus	1.4252	0.490372	0.0026		0.668043	False
swear_liwc22	15.0	23.0	520.0	omnibus	1.4527	0.483674	0.0026		0.668043	False
focusfuture_liwc2015	15.0	23.0	520.0	omnibus	1.1593	0.560087	0.0021		0.746401	False
focusfuture_liwc22	15.0	23.0	520.0	omnibus	1.1113	0.573701	0.002		0.746401	False
affect_liwc2015	15.0	23.0	520.0	omnibus	1.1336	0.567332	0.002		0.746401	False
verb_liwc22	15.0	23.0	520.0	omnibus	1.0907	0.579652	0.002		0.746401	False
verb_liwc2015	15.0	23.0	520.0	omnibus	1.0577	0.58927	0.0019		0.748533	False
affect_liwc22	15.0	23.0	520.0	omnibus	1.0036	0.605446	0.0018		0.758826	False
anx_liwc2015	15.0	23.0	520.0	omnibus	0.6479	0.7233	0.0012		0.894608	False
anx_liwc22	15.0	23.0	520.0	omnibus	0.5776	0.749155	0.001		0.908324	False
compare_liwc2015	15.0	23.0	520.0	omnibus	0.3735	0.829662	0.0007		0.932345	False
auxverb_liwc2015	15.0	23.0	520.0	omnibus	0.3873	0.82394	0.0007		0.932345	False
number_liwc22	15.0	23.0	520.0	omnibus	0.3714	0.830534	0.0007		0.932345	False
compare_liwc22	15.0	23.0	520.0	omnibus	0.3671	0.8323	0.0007		0.932345	False
auxverb_liwc22	15.0	23.0	520.0	omnibus	0.3414	0.843077	0.0006		0.932345	False
number_liwc2015	15.0	23.0	520.0	omnibus	0.3555	0.837145	0.0006		0.932345	False
conj_liwc2015	15.0	23.0	520.0	omnibus	0.2706	0.873461	0.0005		0.94374	False
conj_liwc22	15.0	23.0	520.0	omnibus	0.2813	0.868808	0.0005		0.94374	False
negate_liwc22	15.0	23.0	520.0	omnibus	0.1516	0.926989	0.0003		0.970638	False
negate_liwc2015	15.0	23.0	520.0	omnibus	0.1396	0.932601	0.0003		0.970638	False
ppron_liwc2015	15.0	23.0	520.0	omnibus	0.1521	0.926766	0.0003		0.970638	False
ppron_liwc22	15.0	23.0	520.0	omnibus	0.1245	0.93966	0.0002		0.970638	False
interrog_liwc2015	15.0	23.0	520.0	omnibus	0.013	0.993527	0.0		0.993527	False
interrog_liwc22	15.0	23.0	520.0	omnibus	0.0162	0.991929	0.0		0.993527	False

Post-hoc Pairwise Mann-Whitney U (only BH-surviving omnibus features)
effect_size = rank-biserial correlation; yellow = significant

feature	n_pre	n_during	n_post	comparison	statistic	p_raw	effect_size	note	p_bh	significant
absolutist_liwc2015		23.0	520.0	during_vs_post	5948.5	0.966418	0.0053		0.987427	False
absolutist_liwc2015	15.0	23.0		pre_vs_during	84.0	0.008588	0.513		0.047486	True
absolutist_liwc2015	15.0		520.0	pre_vs_post	1862.0	0.000557	0.5226		0.016734	True
filler_liwc2015		23.0	520.0	during_vs_post	4590.5	0.059245	0.2324		0.192034	False
filler_liwc2015	15.0	23.0		pre_vs_during	126.0	0.16952	0.2696		0.328197	False
filler_liwc2015	15.0		520.0	pre_vs_post	2249.0	0.00517	0.4233		0.030373	True
filler_liwc22		23.0	520.0	during_vs_post	4438.0	0.036306	0.2579		0.16537	False
filler_liwc22	15.0	23.0		pre_vs_during	126.0	0.169497	0.2696		0.328197	False
filler_liwc22	15.0		520.0	pre_vs_post	2247.0	0.005116	0.4238		0.030373	True
informal_liwc2015		23.0	520.0	during_vs_post	3816.0	0.003301	0.3619		0.02387	True
informal_liwc2015	15.0	23.0		pre_vs_during	119.0	0.113468	0.3101		0.248047	False
informal_liwc2015	15.0		520.0	pre_vs_post	1909.0	0.000745	0.5105		0.016734	True
informal_liwc22		23.0	520.0	during_vs_post	3678.0	0.001774	0.3849		0.020863	True
informal_liwc22	15.0	23.0		pre_vs_during	127.0	0.178986	0.2638		0.336494	False
informal_liwc22	15.0		520.0	pre_vs_post	1938.0	0.00089	0.5031		0.016734	True
netspeak_liwc2015		23.0	520.0	during_vs_post	3751.0	0.002474	0.3727		0.020863	True
netspeak_liwc2015	15.0	23.0		pre_vs_during	182.5	0.776515	-0.058		0.923954	False
netspeak_liwc2015	15.0		520.0	pre_vs_post	2693.5	0.041031	0.3094		0.16537	False
netspeak_liwc22		23.0	520.0	during_vs_post	3767.5	0.002663	0.37		0.020863	True
netspeak_liwc22	15.0	23.0		pre_vs_during	183.5	0.753715	-0.0638		0.908324	False
netspeak_liwc22	15.0		520.0	pre_vs_post	2724.0	0.046421	0.3015		0.16537	False

Per-User Rates by Period — Significant Features (BH-corrected)



Section 4 — Sensitivity: Single-Period Users Only

Same Design, Restricted to Users Appearing in Exactly One Period
Removes the between-group dependence of multi-period users; compare significance patterns with Section 3

feature	n_pre	n_during	n_post	comparison	statistic	p_raw	effect_size	note	p_bh	significant
adj_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
adverb_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
affect_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
anger_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
anx_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
article_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
assent_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
auxverb_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
compare_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
conj_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
filler_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
focusfuture_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
focuspast_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
focuspresent_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
function_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
i_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
informal_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
interrog_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
ipron_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
negate_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
negemo_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
netspeak_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
nonflu_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
number_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
posemo_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
ppron_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
prep_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
pronoun_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
quant_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
sad_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
shehe_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
swear_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
they_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
verb_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
we_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
you_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
absolutist_liwc2015	3	2	503	omnibus				skipped: <2 periods with ≥5		False
function_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
pronoun_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
ppron_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
i_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
we_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
you_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
shehe_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
they_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
ipron_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
article_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
prep_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
auxverb_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
adverb_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
conj_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
negate_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
verb_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
adj_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
compare_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
interrog_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
number_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
quant_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
affect_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
posemo_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
negemo_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
anx_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
anger_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
sad_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
focuspast_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
focuspresent_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
focusfuture_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
informal_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
swear_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
netspeak_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
assent_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
nonflu_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False
filler_liwc22	3	2	503	omnibus				skipped: <2 periods with ≥5		False